Computer Programming II (CSE2035)
Week 3 Lab Session:
C String Processing

Prof. Jaeseung Choi
Dept. of Computer Science and Engineering
Sogang University

Environment

- **We will use Linux-style environment to compile and execute C/C++ programs**
  - Ex) Linux virtual machine, WSL, MacOS, Cygwin, **MinGW**
  - You should be able to run the commands like `make`, `gcc`, `g++` ...
  - This is to grade your assignments easily

- **You can choose your favorite editor to write code**
  - Ex) vim, emacs, Visual Studio Community, **Visual Studio Code**

- **We provide MinGW + Visual Studio Code in Lab PCs**
  - If you want to install these in your own laptop, check the manual uploaded in the cyber campus

Lab Session Outline

- **I will give you 2~4 problems each week**
  - Specification and skeleton code will be provided
  - You can compile the program by typing "`make`" in the terminal
  - You have to fill in empty function(s) according to the spec
- **These problems are (1) in-class activities and (2) your assignment at the same time**
  - TA will stay in the class to answer your questions
  - If you solve all the problems early, you can submit them and leave the class starting from 10:00
- **Submission guideline will be provided at the end**

Exercise - Computing String Length

- **Write a function that returns the length of the input string ‘s’**
- **Function Prototype**

```
size_t my_strlen(char *s);
```

- **Assumption**
  - `s` is a ***valid string***: this means that (1) `s` not a NULL pointer and (2) the string in `s` is properly terminated with a null character (`'\0'`)
- **Constraint**
  - Do not use the existing `strlen( )` function from the C library

Before we look into the skeleton code…

- **Command-line argument**
  - In `main()`, you can access the strings that are passed as the command-line arguments of the program

Source code

```c
int main(int argc, char ** argv) {
    printf("%d arguments\n", argc);
    for (int i = 0; i < argc; i++)
        printf("Arg %d = %s\n", i, argv[i]);
    return 0;
}
```

Execution result

```
$ ./test.bin hello world
3 arguments
Arg 0 = ./test.bin
Arg 1 = hello
Arg 2 = world
```

Demonstration

- First, visit our cyber campus and find “Week 3 Lab Assignment” page
- Download the attached “week3.zip” file

Homework 3-1: Reversing Strings

- **Write a function that reverses the input string ‘s’. This means that the order of characters in the string must be reversed.**
- **Function Prototype**

```
void reverse(char *s);
```

- **Assumption**
  - s is a valid string
- **Example**

```
char str[16] = "abcd";
reverse(str);
printf("%s\n", str); // Should print "dcba\n"
```

Homework 3-2: Safe String Copy

- **Let's write a safe version of `strcpy()`. This function copies string `src` into the char array `dst` that has size `sz`.**
- **Function Prototype**

```
void safe_strcpy(char *dst, char *src, size_t sz);
```

- **Assumption**
  - `src` is a valid string
  - `dst` is a character array whose size is `sz`
- **Spec**
  - After this function is executed, `dst` must be null-terminated
  - This function should not write past the end of `dst` (= no overflow!)
    - Therefore, if the length of `src` is greater than or equal to `sz`, then you should copy just `sz-1` chars
- **Note that this behavior is different from `strncpy()`**

Homework 3-3: Safe String Concat

- **Let's write a safe version of `strcat()`. This function copies string `src` and appends it after the string in `dst`.**
- **Function Prototype**

```
void safe_strcat(char *dst, char *src, size_t sz);
```

- **Assumption**
  - `src` is a valid string
  - `dst` is a character array whose size is `sz`
  - `dst` contains a string whose length is less than `sz`
- **Spec**
  - After this function is executed, `dst` must be null-terminated
  - This function should not write past the end of `dst` (= no overflow!)
- **Note that this behavior is different from `strncat()`**

Submission Guideline

- **Submit the completed `hw3-1.c`, `hw3-2.c`, `hw3-3.c` files to the "Week 3 Lab Assignment" page in cyber campus**
  - Do not zip these files, just upload multiple files in the form
  - If your submission format is wrong, 10% deduction
  - If the submitted file doesn't compile with the "`make`" command, you will get 0 point

- ~~**Deadline is 9/21 (Wednesday) 23:59**~~
  - Deadline is postponed to 9/28 (Wednesday) 23:59

- ~~**Late submission is allowed until 9/23 (Friday) 23:59**~~
  - Late submission deadline is postponed to 9/30 (Friday) 23:59
  - 20% deduction

Q & A Session